Cut Sheet

Board Design Summary

Style: tumbling-block-3d
Dimensions: 12" (305 mm) × 18" (457 mm) × 1-1/2" (38 mm)
Species: Maple, Cherry, Walnut

Cut List

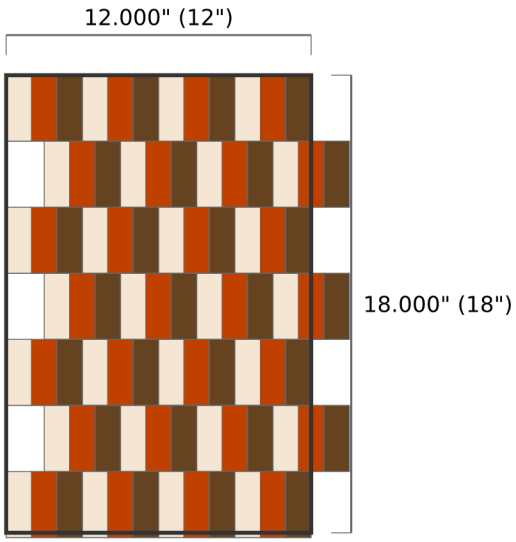
Part	Width	Thickness	Length	Species
Board strips	1.000" (1")	1.500" (1 1/2")	18.000" (18")	Maple

Cut List

Part	Qty	Width	Thickness	Length	Species
Maple (light) diamond strips: diamond side 1-1/2" (38 mm), 60°/120° angles; mill stock 1-5/16" (33 mm) thick (side × sin 60°) and at least 6" (152 mm) wide; bevel the wide stock at 30° first, then cut each diamond off with a thin-rip jig, keeper away from the fence; 6 strips per tone × 20-3/16" (513 mm) long	6	6" (152 mm)	1-5/16" (33 mm)	20-3/16" (513 mm)	Maple
Cherry (mid) diamond strips: diamond side 1-1/2" (38 mm), 60°/120° angles; mill stock 1-5/16" (33 mm) thick (side × sin 60°) and at least 6" (152 mm) wide; bevel the wide stock at 30° first, then cut each diamond off with a thin-rip jig, keeper away from the fence; 6 strips per tone × 20-3/16" (513 mm) long	6	6" (152 mm)	1-5/16" (33 mm)	20-3/16" (513 mm)	Cherry

Walnut (dark) diamond strips: diamond side 1-1/2" (38 mm), 60°/120° angles; mill stock 1-5/16" (33 mm) thick (side × sin 60°) and at least 6" (152 mm) wide; bevel the wide stock at 30° first, then cut each diamond off with a thin-rip jig, keeper away from the fence; 6 strips per tone × 20-3/16" (513 mm) long	6	6" (152 mm)	1-5/16" (33 mm)	20-3/16" (513 mm)	Walnut
Log A: one hexagon row wide, 6 hexagons side by side (one diamond strip of each tone, Maple/Cherry/Walnut, per hexagon) glued in a 60° cradle or by the masking-tape roll-up method, light clamp pressure only; logs are never planed or jointed (15-19/32" (396 mm) wide: too wide for the planer at 15-19/32" (396 mm) and the jointer; crosscut only on the sled); log 20-3/16" (513 mm) long; crosscut 10 slices × 1-11/16" (43 mm) each on a crosscut sled with a stop block (9 rows used + 1 spare: slice 10 spare; 6 hexagons per slice); offset alternate rows by half a hexagon, 1-5/16" (33 mm)	1	15-19/32" (396 mm)	2-19/32" (66 mm)	20-3/16" (513 mm)	Maple/Cherry/ Walnut

Board Layout (Top View)



Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3.2 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

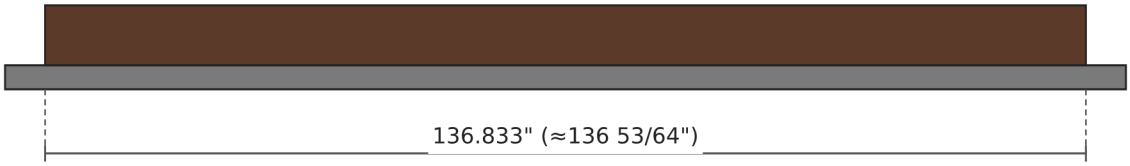
Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.7 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

3D Tumbling Block (End Grain): step 1: Mill rough stock

Whole blank on the jointer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 271.697" (≈271 45/64") long, 56 slices of 4.692" (≈4 11/16") crosscut after milling.

Maple strip: blank 136.833" (≈136 53/64") long, 28 slices of 4.692" (≈4 11/16") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 2: Mill tone boards

Mill one board per tone (light Maple, mid Cherry, dark Walnut) to 1-5/16" (33 mm) thick (diamond side 1-1/2" (38 mm) × sin 60°) and at least 6" (152 mm) wide, 20-3/16" (513 mm) long. Rip nothing narrow yet.

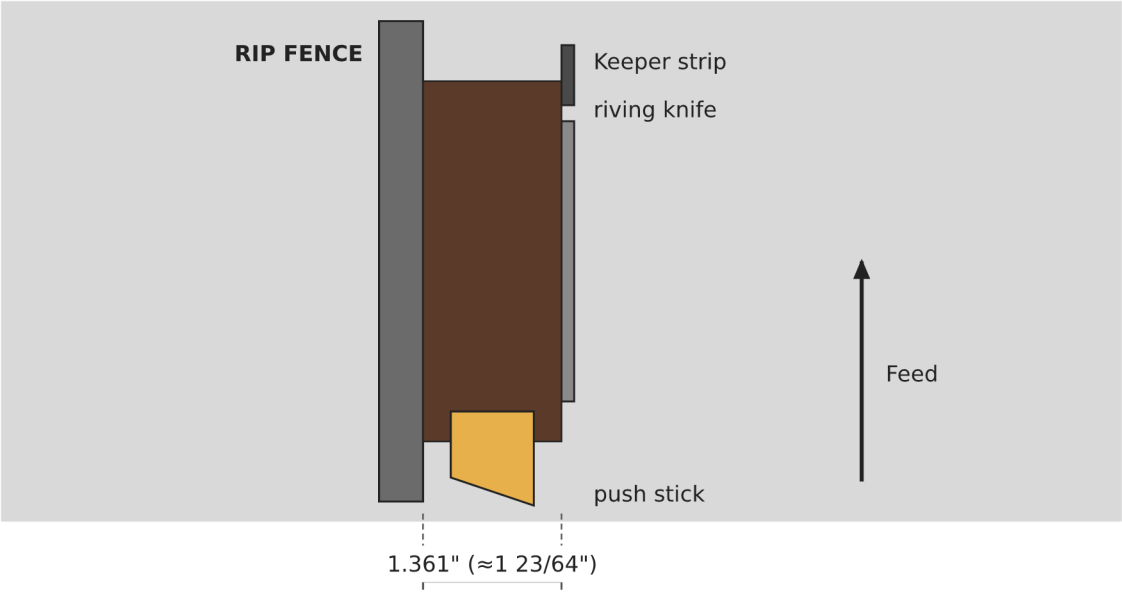
Settings:

Fence to blade: thin-rip jig set to the diamond side 1-1/2" (38 mm)

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

3D Tumbling Block (End Grain): rip-fence setup, step 2: Rip strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade

Rip three species to equal width with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

Step 3: Bevel strips

Bevel the wide stock first: tilt the blade 30° and bevel one edge of each wide board. Then cut each 60°/120° diamond (side 1-1/2" (38 mm)) off with a thin-rip jig, the keeper (the diamond) falling away from the fence: the thin-rip jig and the keeper go on the side the blade tilts away from: on a right-tilt saw, the jig and keeper are left of the blade. Re-bevel the wide board's edge before each next diamond. Never bevel a strip that is already narrow. 6 diamond strips per tone.

Settings:

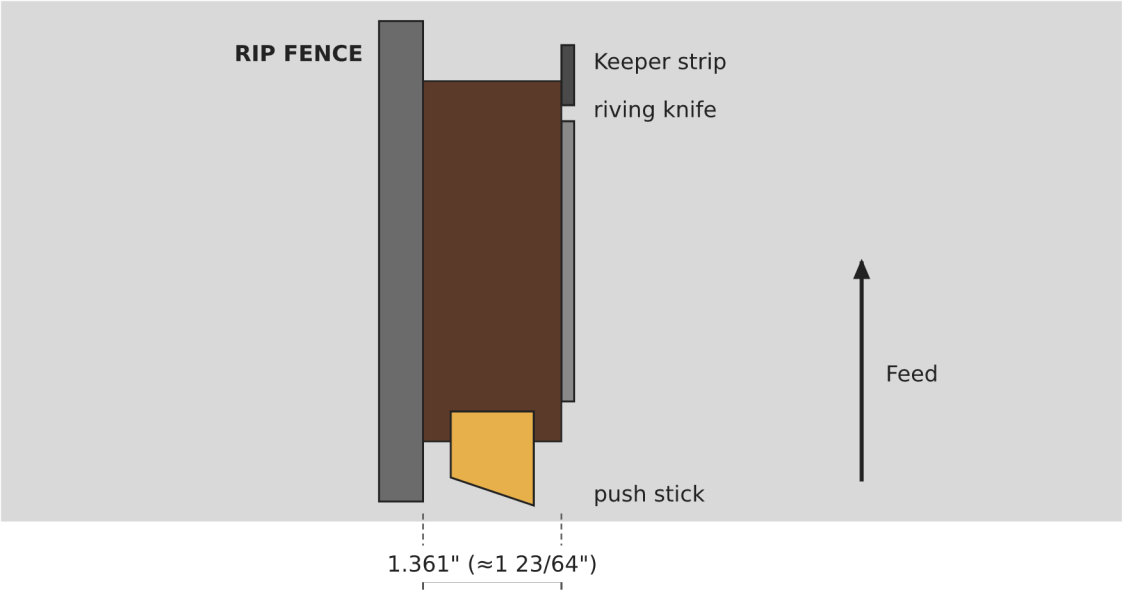
Blade tilt: 30°

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

WARNING: Featherboard holding the stock down and against the jig, push block for the last 12", keeper on the jig side away from the fence, never trap a piece between the blade and the fence.

3D Tumbling Block (End Grain): rip-fence setup, step 3: Bevel strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade

Rip a 30° bevel on both edges to make 60° diamonds in section. Use a hold-down; the offcut side must be the outside of the blade.

Blade tilt: 30°

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

WARNING: Make angled cuts on a sled or miter gauge with a hold-down, never against the rip fence alone.

Step 4: Glue up

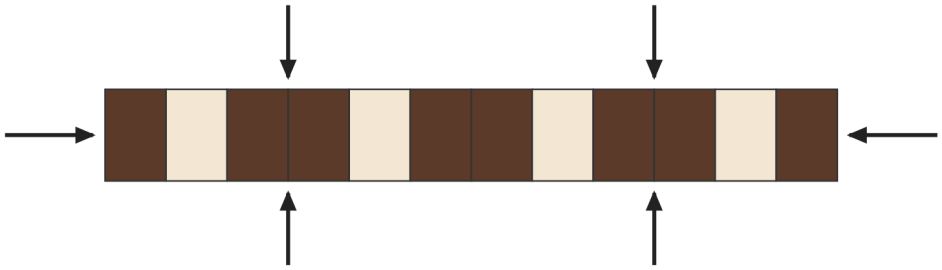
Glue the diamond strips into one log one hexagon row wide: 6 hexagons side by side, one diamond strip of each tone per hexagon, 20-3/16" (513 mm) long, in a 60° cradle or by the masking-tape roll-up method, light clamp pressure only. Logs are never planed.

Settings:

Clamping: light, even pressure in the 60° cradle or with tape; no bar-clamp spacing.

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

3D Tumbling Block (End Grain): glue-up 1, step 4: Glue up



Glue the diamonds into hexagon logs, one of each tone. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

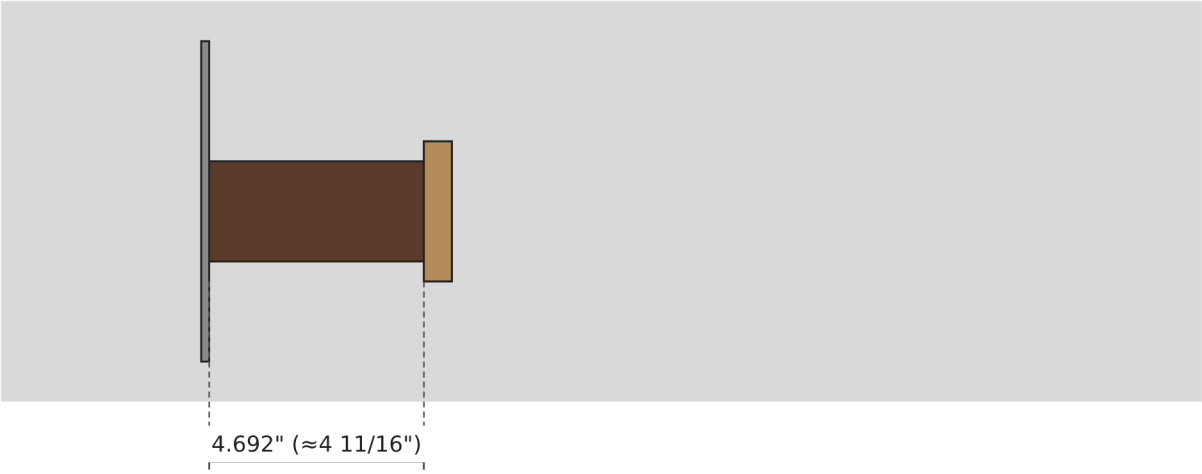
Step 5: Crosscut slices

Crosscut the log into 10 slices on a crosscut sled with a stop block; each slice is a whole row strip of 6 hexagons.
Cut height: 2-19/32" (66 mm) (log height). Blade height: set the blade 1/8" (3.2 mm) above the log, 2-23/32" (69 mm).
If a slice is not cut through in one pass, finish it on the bandsaw; never flip the log for a second pass on the sled.
Slice size: 1-11/16" (43 mm).
Final thickness: 1-1/2" (38 mm).
End-grain allowance: 3/16" (5.0 mm).
Slices: 10 in all (Log A 10).
Sled width: sled must clear the log, 15-19/32" (396 mm).

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

3D Tumbling Block (End Grain): crosscut, step 5: Crosscut slices



Crosscut the logs into slices on a crosscut sled with a stop block.

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Step 6: Glue up

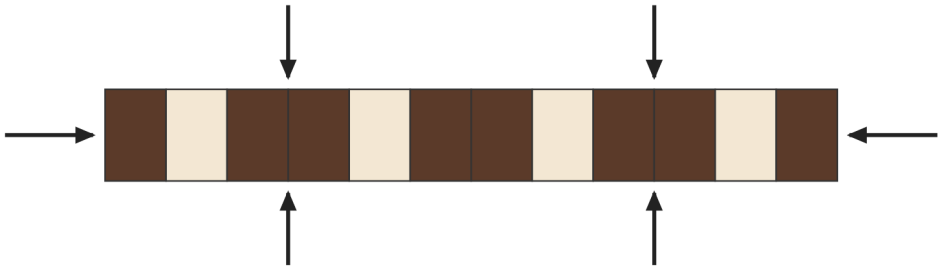
Nest the row strips (slices) with alternate rows offset by half a hexagon, 1-5/16" (33 mm). Glue them in pairs, then join the pairs, on a flat caul with light, even pressure. Trim the zig-zag edges on the sled with a stop block and hold-down. 9 rows used, 1 spare.

Settings:

Clamp spacing: about 8" (203 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

3D Tumbling Block (End Grain): glue-up 2, step 6: Glue up



Glue hexagon slices into the final panel, then trim the zig-zag edges square. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Step 7: Flatten and sand

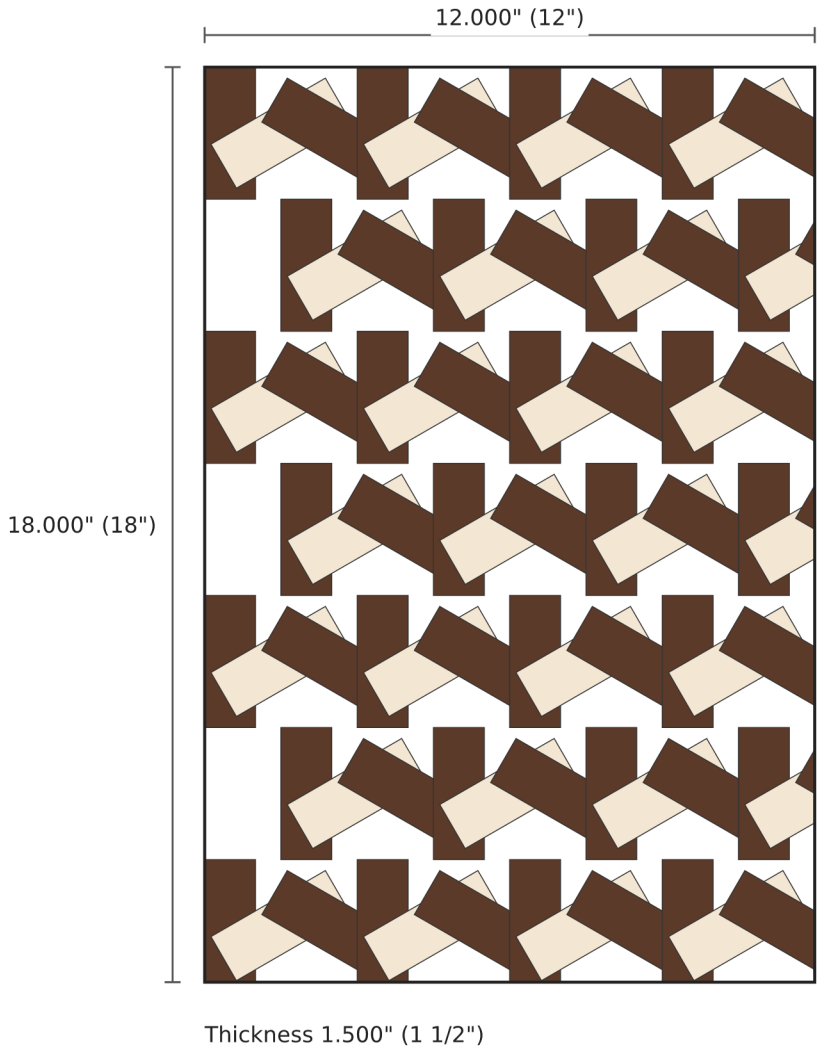
Flatten the final glue-up, 12" (305 mm) wide:
mixed grain: router sled or drum sander, never the planer.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

3D Tumbling Block (End Grain): top view



Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Step 8: Edge profile

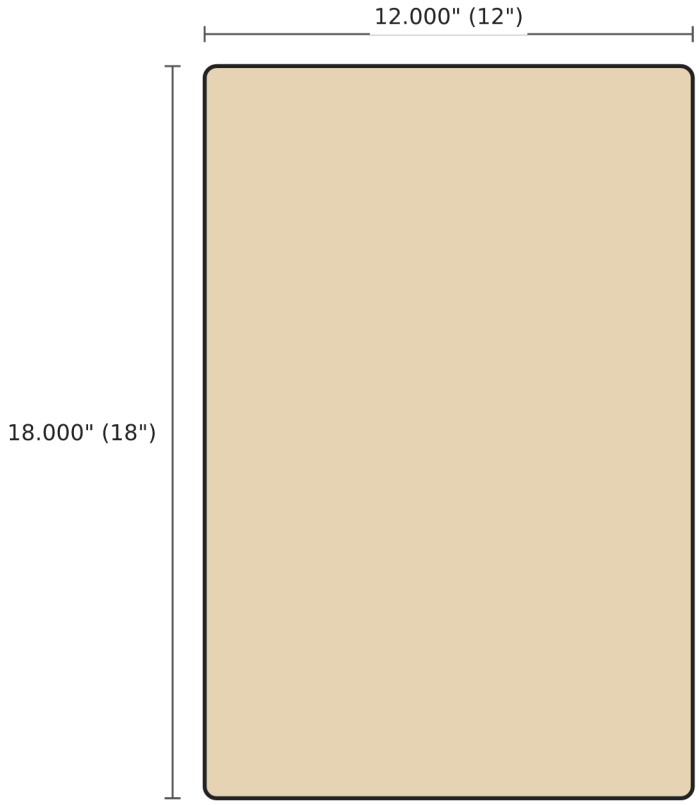
Rout the edge profile, default: 1/8" (3.2 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

Settings:

Bit: 1/8" (3.2 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

3D Tumbling Block (End Grain): outline, step 8: Edge profile



Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 9: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.